



ISLAMIC UNIVERSITY OF TECHNOLOGY

Department of Electrical & Electronic Engineering

Course No.: EEE 4418

Course Name: Electrical Services Design Lab

AutoCAD Project

Section: A2

Group: 2

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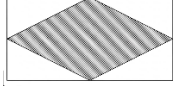
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Introduction:

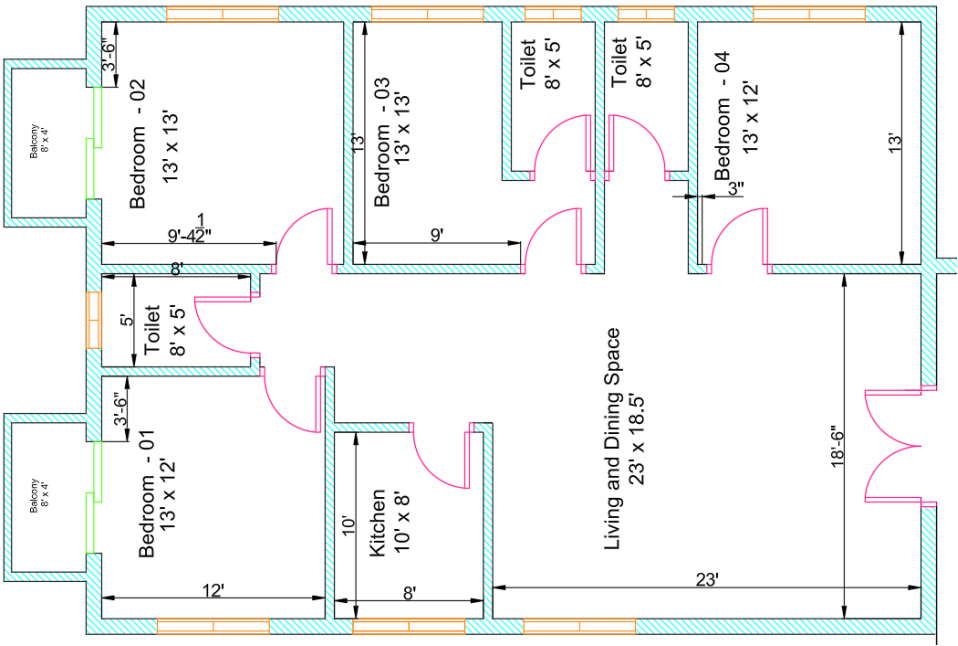
This project shows the complete design of a seven-story building, along with all the floor plans (ground, floors and rooftop) and electrical layout being displayed in detail. All the calculations are also shown. This is a residential building, with 6 floors, and 4 units per floor. The building is symmetric in the horizontal and the vertical direction.

The ratings of the wires are also shown along with the electrical layout. We have also shown the circuit diagram (single-line) of the Main Distribution Board (MDB) and the Emergency Main Distribution Board (EMDB). AutoCAD 2027 (Student's Access) has been used for designing all the circuits, the conduits layouts, and the floor designs.

The table below shows the symbols of the electrical appliances and fixtures along with their power rating:

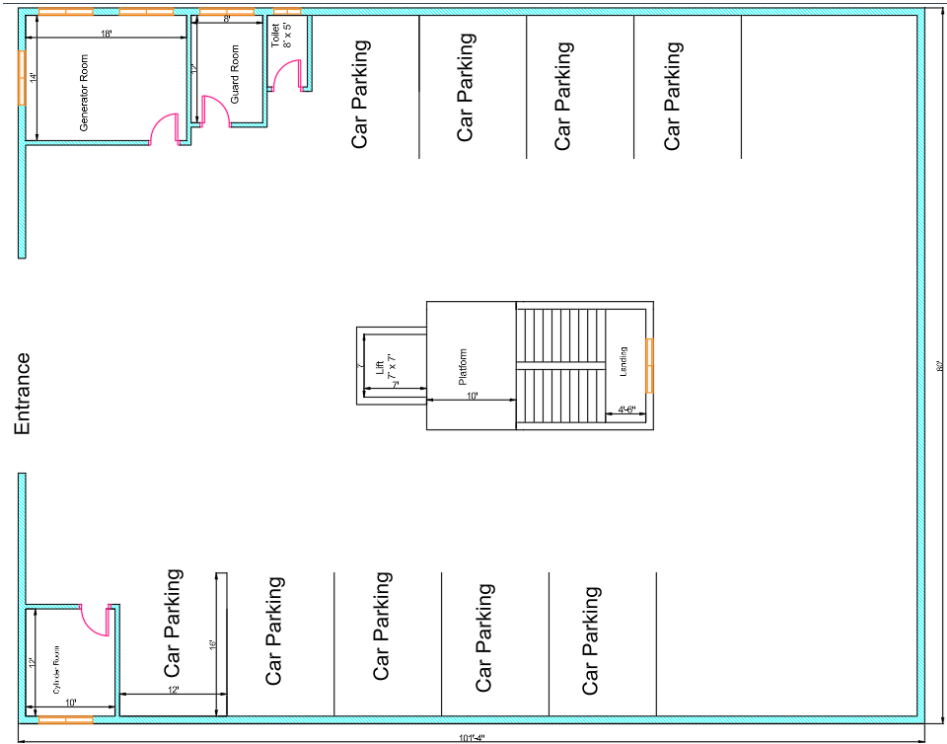
Symbol of the Appliance / Fixture	Legend	Description	Power Rating (W)
	F	Ceiling Fan	100
	L	Bracket-Mounted Light Bulb	20
	CL	Ceiling-Mounted Light Bulb	20
	BDB	Branch Distribution Board	—
	MDB	Main Distribution Board	—
	EMDB	Emergency Main Distribution Board	—
	EBDB	Emergency Branch Distribution Board	—
	SB	Switchboard	—

One Unit in Broad

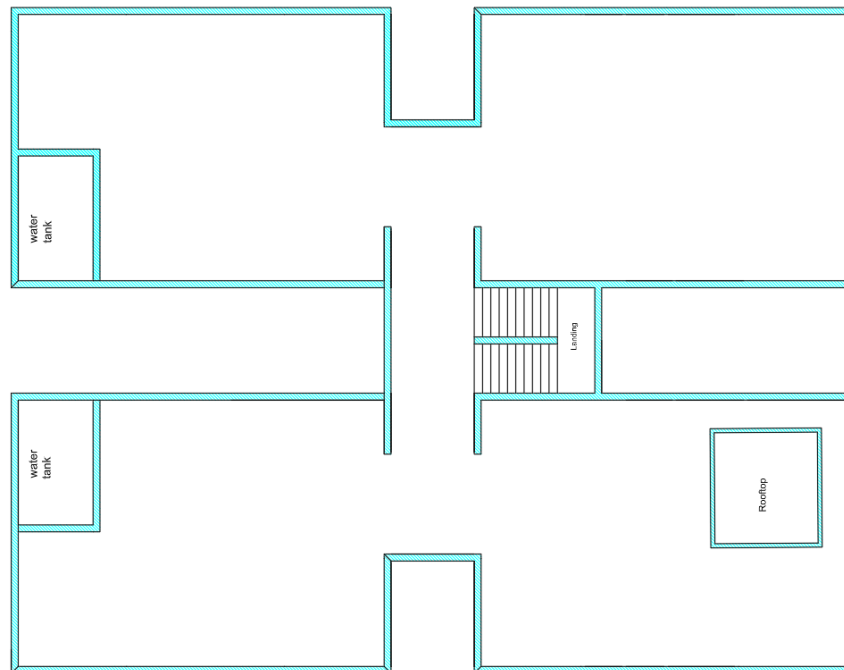


This is one unit. It has 4 bedrooms, 3 toilets, one kitchen, one living and dining space, along with two balconies. Since the building is symmetric, all the other units follow the same arrangement.

Garage and Ground Floor:



Rooftop:



Electrical Fittings and Fixtures:

The fixtures and fittings calculations are shown below:

For Lights,

Illuminance, $E = 100 \text{ lumen/m}^2$

Light Loss Factor and Utilization Factor, $LLF \times UF = 0.7 \times 0.77$

Number of Lights per Illumination, $n = 1$

Flux = 2000 lumen Area = A

Number of Lights = $(E \times A) / (n \times F \times UF \times LLF)$

Number of fans = $A / 150$

For Bedroom 1,

Area = $13 \times 12 = 156 \text{ ft}^2$

Area = $156 \times 0.092903 = 14.49 \text{ m}^2$

Number of Lights = $(100 \times 14.49) / (1 \times 2000 \times 0.7 \times 0.77) = 1.34 \sim 2$

Number of Fans = $156 / 150 = 1.04 \sim 1$

(It must be noted that since our room is almost square shaped, placing two fans put them too close together, so we are using one fan so they don't overlap)

For Bedroom 2,

Area = $13 \times 13 = 169 \text{ ft}^2$

$$\text{Area} = 169 \times 0.092903 = 15.70 \text{ m}^2$$

$$\text{Number of Lights} = (100 \times 15.70) / (1 \times 2000 \times 0.7 \times 0.77) = 1.46 \sim 2$$

$$\text{Number of Fans} = 169 / 150 = 1.127 \sim 1$$

For Bedroom 3,

$$\text{Area} = 13 \times 13 = 169 \text{ ft}^2$$

$$\text{Area} = 169 \times 0.092903 = 15.70 \text{ m}^2$$

$$\text{Number of Lights} = (100 \times 15.70) / (1 \times 2000 \times 0.7 \times 0.77) = 1.46 \sim 2$$

$$\text{Number of Fans} = 169 / 150 = 1.127 \sim 1$$

For Bedroom 4,

$$\text{Area} = 13 \times 12 = 156 \text{ ft}^2$$

$$\text{Area} = 156 \times 0.092903 = 14.49 \text{ m}^2$$

$$\text{Number of Lights} = (100 \times 14.49) / (1 \times 2000 \times 0.7 \times 0.77) = 1.34 \sim 2$$

$$\text{Number of Fans} = 156 / 150 = 1.040 \sim 1$$

For Kitchen,

$$\text{Area} = 10 \times 8 = 80 \text{ ft}^2$$

$$\text{Area} = 80 \times 0.092903 = 7.43 \text{ m}^2$$

$$\text{Number of Lights} = (100 \times 7.43) / (1 \times 2000 \times 0.7 \times 0.77) = 0.69 \sim 1$$

$$\text{Number of Fans} = 80 / 150 = 0.533 \sim 1$$

For living and Dining Space,

$$\text{Area} = 23 \times 18.5 = 425.5 \text{ ft}^2$$

$$\text{Area} = 425.5 \times 0.092903 = 39.53 \text{ m}^2$$

For dining room, we are using less bright, 1250 Lumens, 20 W bulbs:

$$\text{Number of Lights} = (100 \times 39.53) / (1 \times 1250 \times 0.7 \times 0.77) = 5.87 \sim 6$$

$$\text{Number of Fans} = 425.5 / 150 = 2.837 \sim 3$$

For Toilet 1,

$$\text{Area} = 8 \times 5 = 40 \text{ ft}^2$$

$$\text{Area} = 40 \times 0.092903 = 3.72 \text{ m}^2$$

$$\text{Number of Lights} = (100 \times 3.72) / (1 \times 1250 \times 0.7) = 0.425 \sim 1$$

For Toilet 2,

$$\text{Area} = 8 \times 5 = 40 \text{ ft}^2$$

$$\text{Area} = 40 \times 0.092903 = 3.72 \text{ m}^2$$

$$\text{Number of Lights} = (100 \times 3.72) / (1 \times 1250 \times 0.7) = 0.425 \sim 1$$

For Toilet 3,

$$\text{Area} = 8 \times 5 = 40 \text{ ft}^2$$

$$\text{Area} = 40 \times 0.092903 = 3.72 \text{ m}^2$$

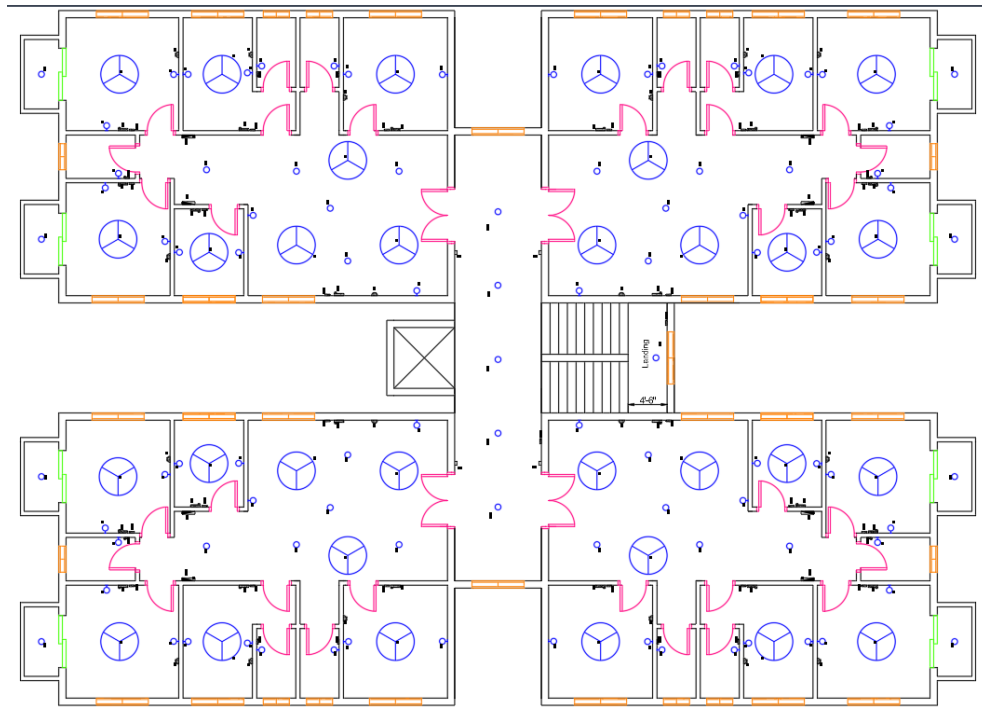
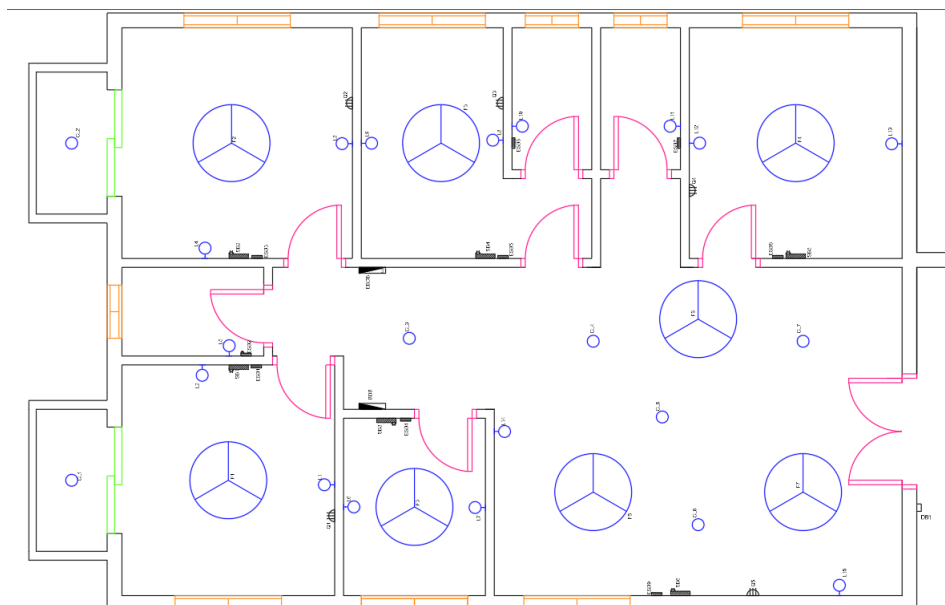
$$\text{Number of Lights} = (100 \times 3.72) / (1 \times 1250 \times 0.7) = 0.425 \sim 1$$

For Balcony,

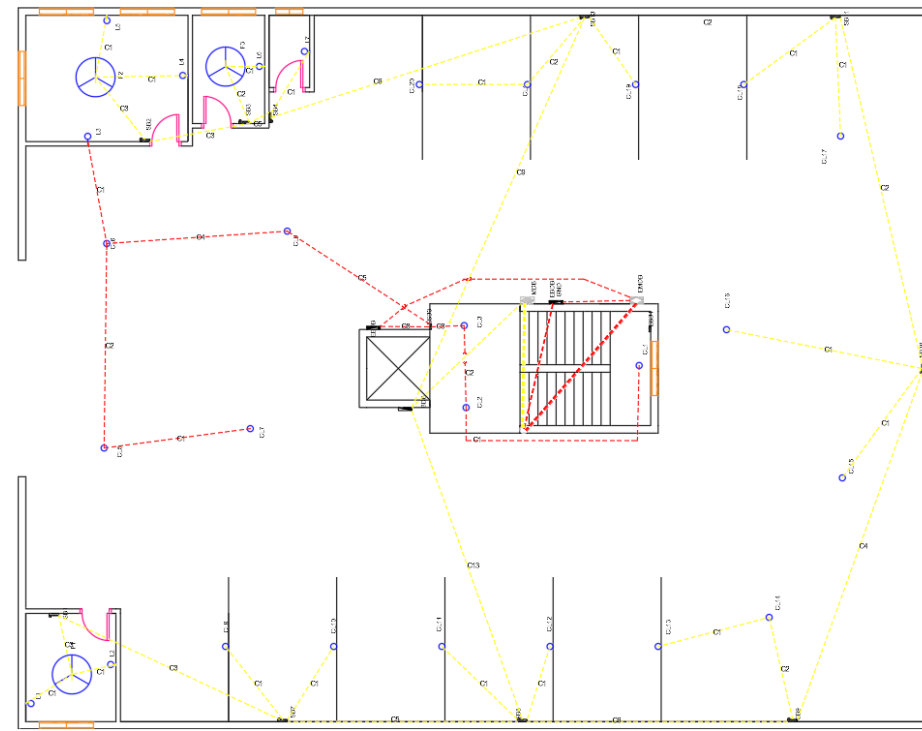
Number of Lights = 2, one in each balcony

The balcony lights are taken from the lighting points shown in the floor plan.

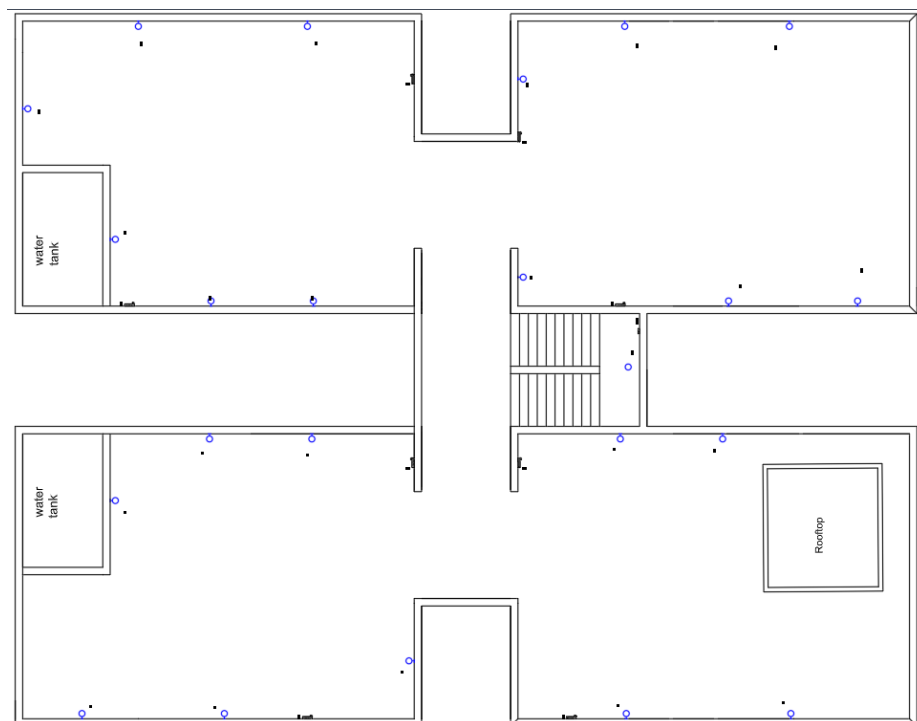
Finally, the fitting and fixtures are properly indicated below:

Typical Floor:**One Unit in Detail:**

Garage and Ground Floor:

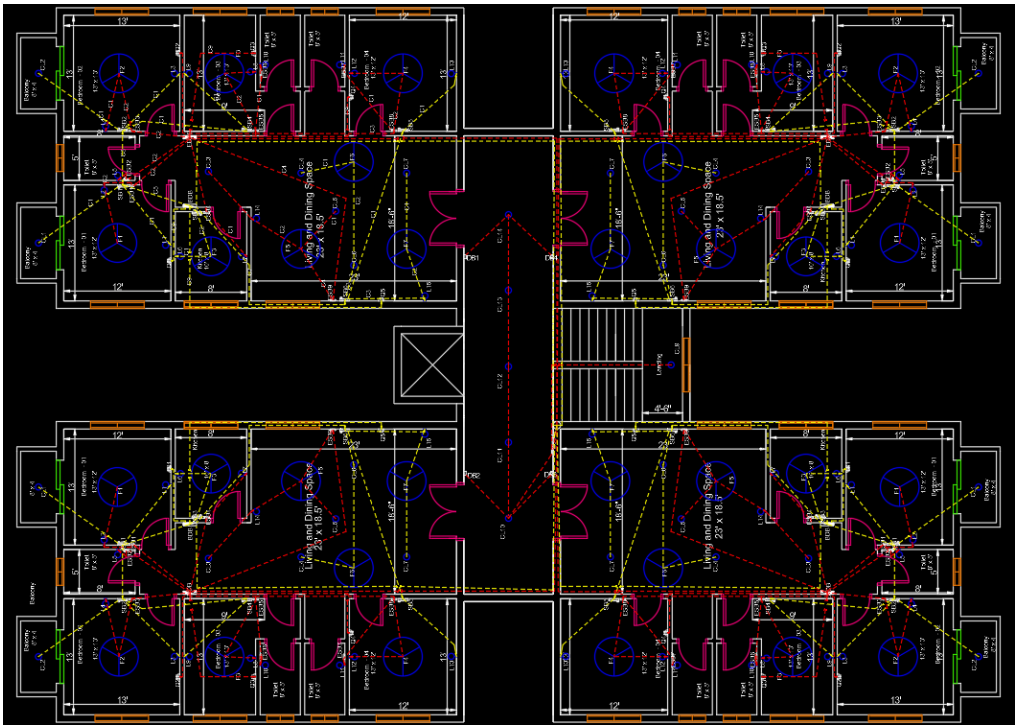


Rooftop:

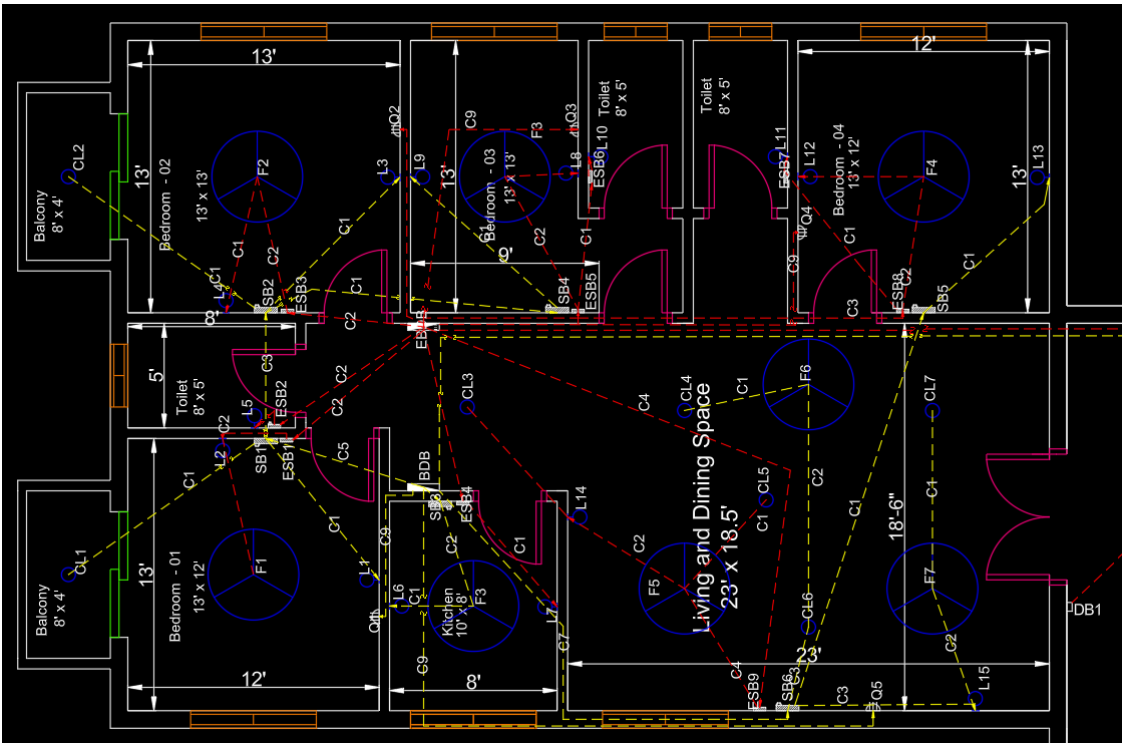


The conduit layout is shown below:

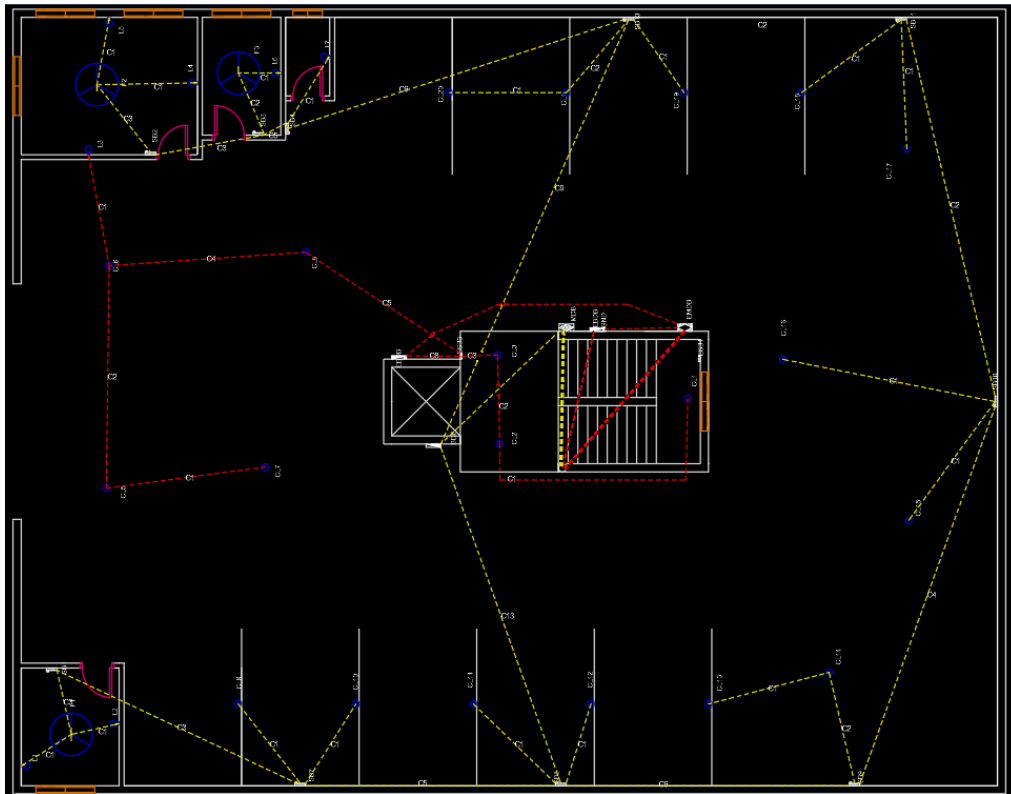
Typical Floor (in dark mode so that it's easier to understand the yellow conduits):



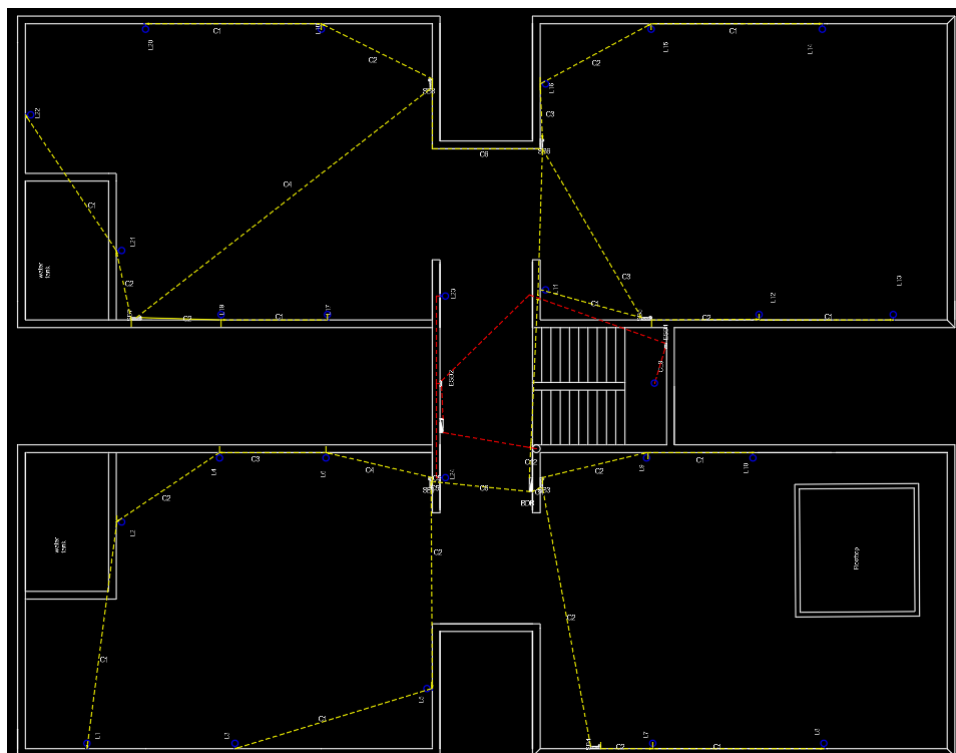
One Unit in Detail:



Garage and Ground Floor:



Rooftop:



Load Calculations and Conduit Selection:

The following table has been used to select the cables:

A	B	C	D	E	F		G	H	I		J	
					a'	b'			a''	b''	a'''	b'''
3/0.029	1.5	5	16	10	6	10		27	27	22	16	20
7/0.029	2.5	10	16	10	4	7		16	36	30	22	28
7/0.036	4	15	14	10	3	5	1	10	47	39	30	37
7/0.044	6	20	14	10	2	4	1	6.8	59	50	38	47
7/0.052	10	30	10	10	1	2	1.5	4	78	68	52	63
7/0.064	16	40	10	10		1	1.5	2.6	100	94	70	85
19/0.052	25	50	6	6		1	2	1.6	130	125	91	110
19/0.064	35	60	6	6			2	1.2	155	160	112	136
19/0.072	50	70	6	6			2	0.93	185	195	136	164
19/0.083	70	100	1/0	1/0			2	0.65	225	245	173	207
37/0.072	95	120	1/0	1/0			2.5	0.48	270	300	216	253
37/0.083	120	150	1/0	1/0			2.5	0.4	310	350	244	291
37/0.093	150	200	1/0	1/0			3	0.34	350	405		333
37/0.130	185	250	3/0	3/0			3.5	0.29	390	460		381
61/0.093	240	300	3/0	3/0			4	0.24	450	555		452
61/0.103	300	425	3/0	3/0			4	0.22	515	640		526
91/0.093	400	585	3/0	3/0			6	0.2	586	770		639
91/0.103	500	685	3/0	3/0			6	0.18	680	900		752
127/0.103	630	800	3/0	3/0			6	0.17	800	1030		855

The conduits have been named using this table (both tables are from the lab manual):

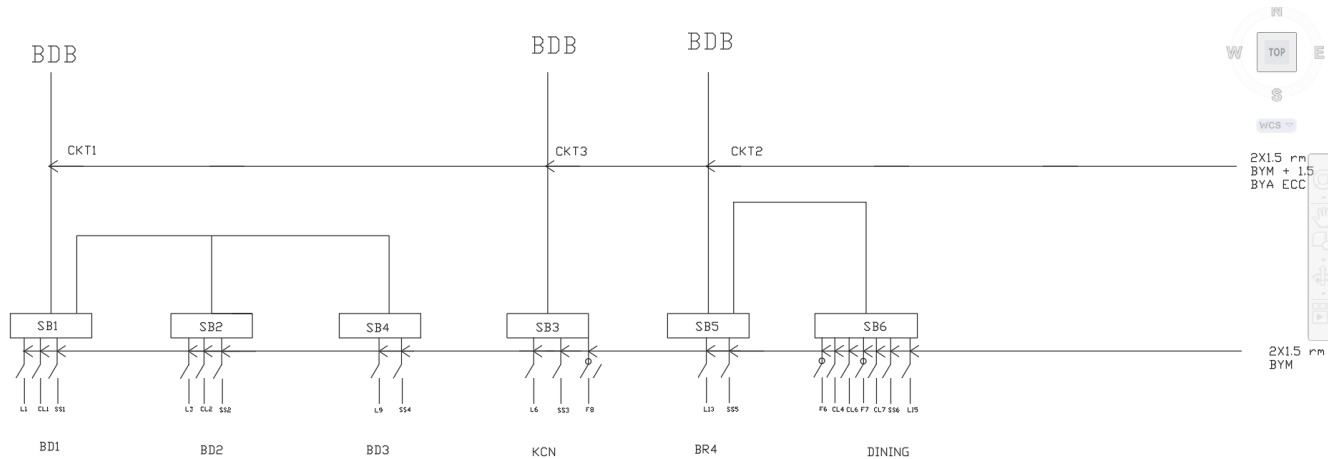
C1	2 x 1.5 mm ²
C2	4 x 1.5 mm ²
C3	6 x 1.5 mm ²
C4	8 x 1.5 mm ²
C5	10 x 1.5 mm ²
C6	12 x 1.5 mm ²
C7	14 x 1.5 mm ²
C8	2 x 2.5 mm ²
C9	2 x 4 mm ²
C8,9	2 x 2.5 mm ² and 2 x 4 mm ²

The wires have already been labelled in the conduit layout, which is shown in the previous pages.

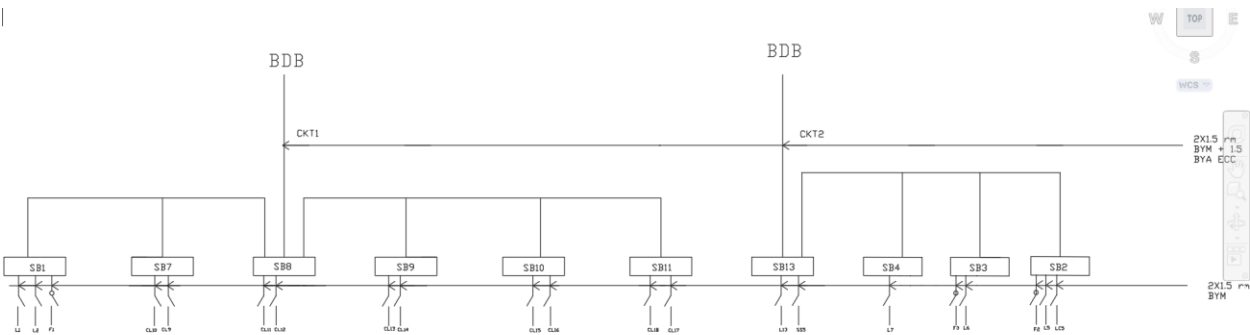
Distribution Board and Single Line Diagram:

The diagrams are shown below, and the necessary load calculations used are shown in the next section

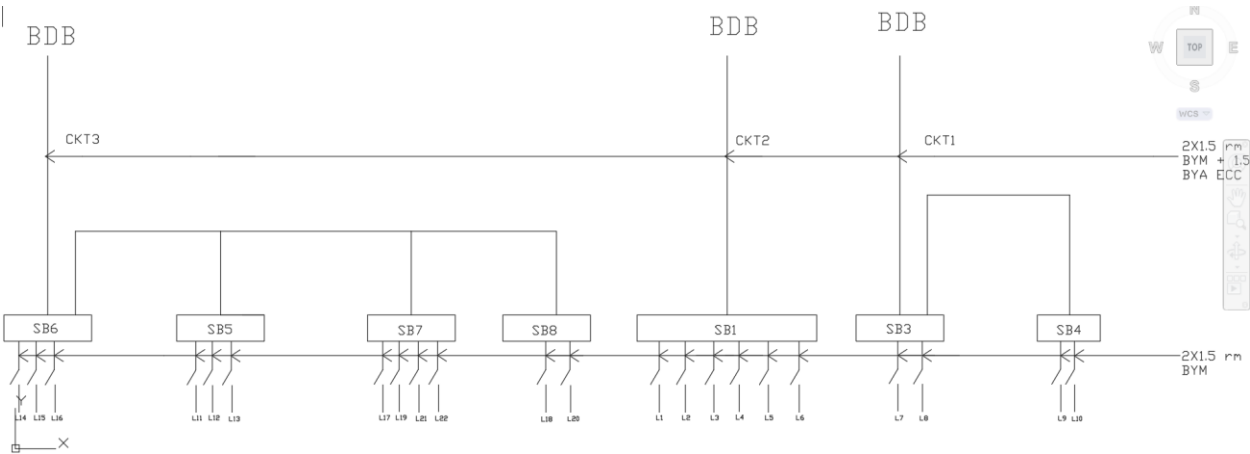
Floor BDB Switchboard Circuits:



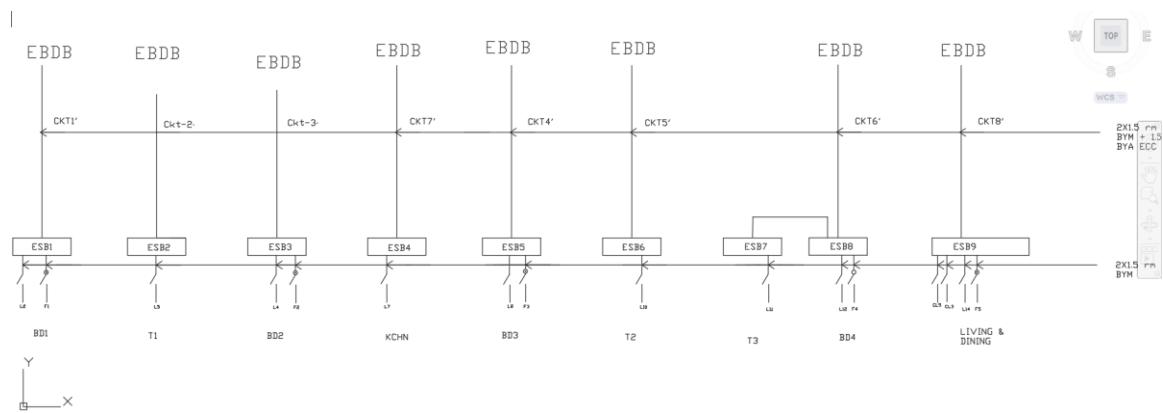
Ground BDB Switchboard Circuits:



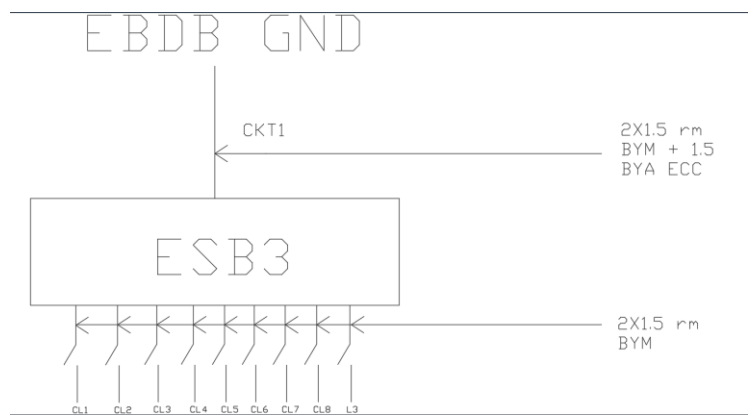
Rooftop BDB Switchboard Circuits:



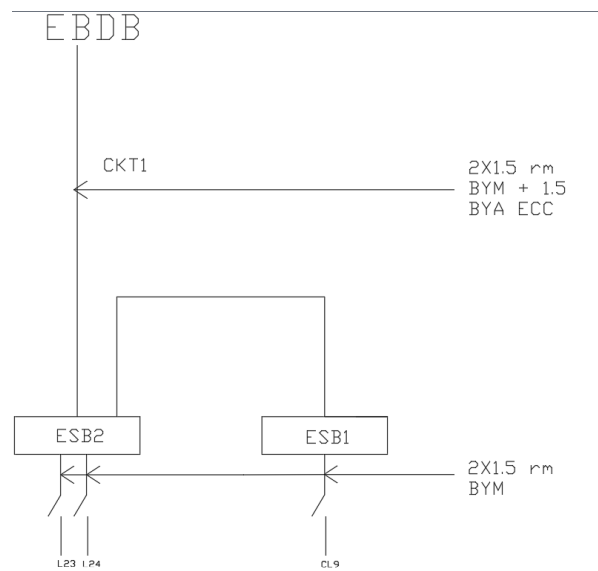
Floor EBDB Switchboard Circuits:



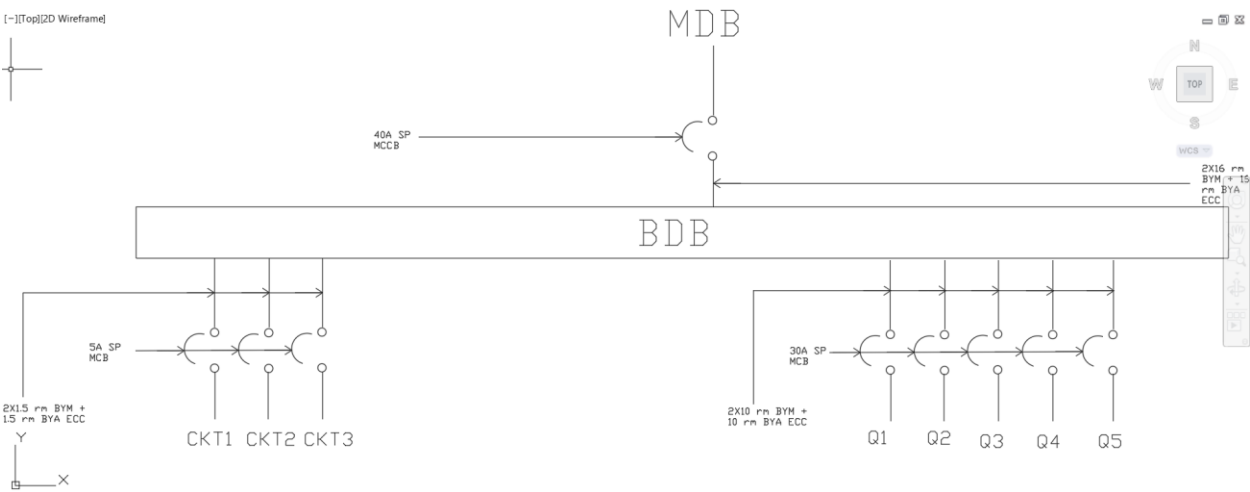
Ground EBDB Switchboard Circuits:



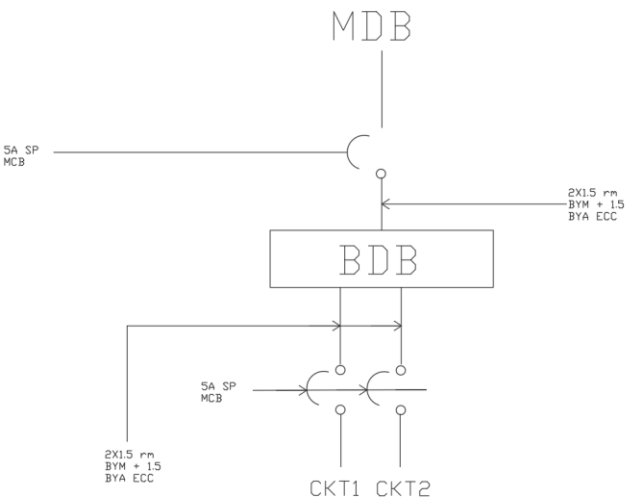
Rooftop EBDB Switchboard Circuits:



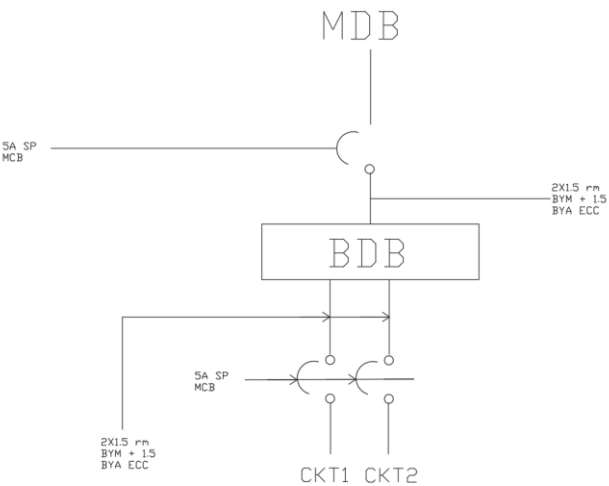
BDB Diagram For Each Unit:



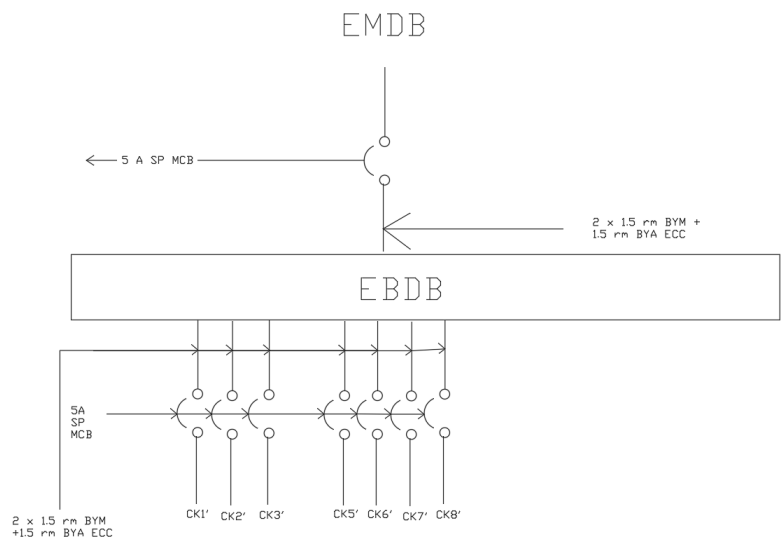
BDB Diagram for Ground:



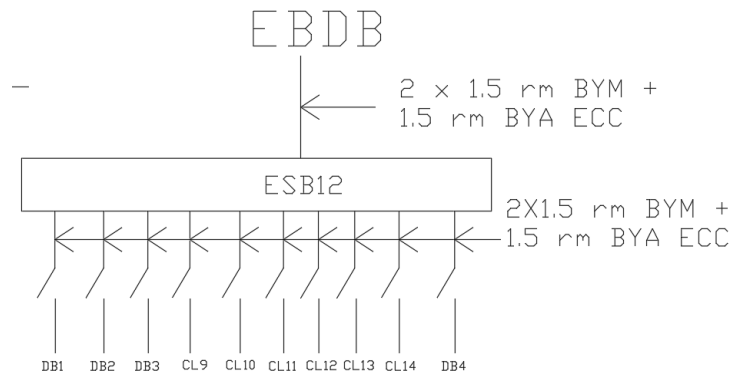
BDB Diagram for Rooftop:



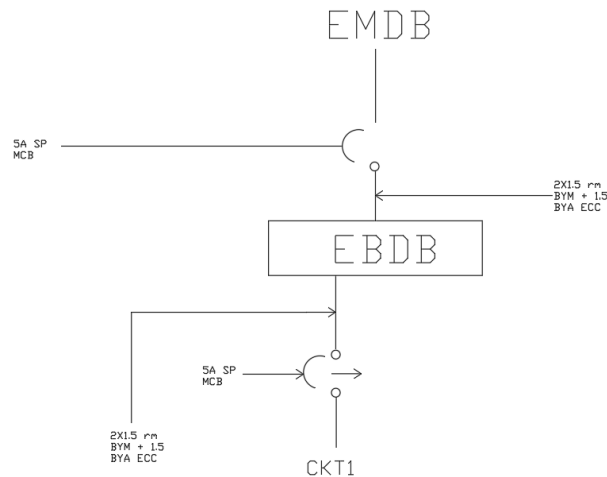
EBDB Diagram For Each Unit:



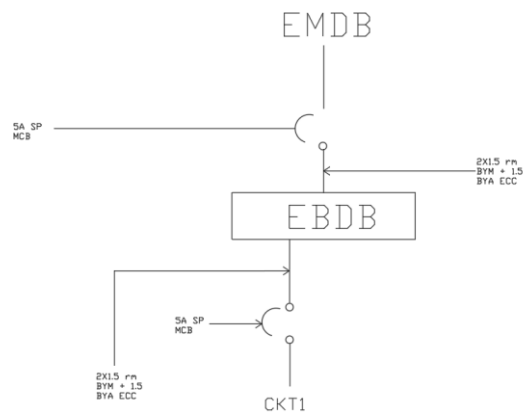
EBDB Diagram for Floor Lobby:



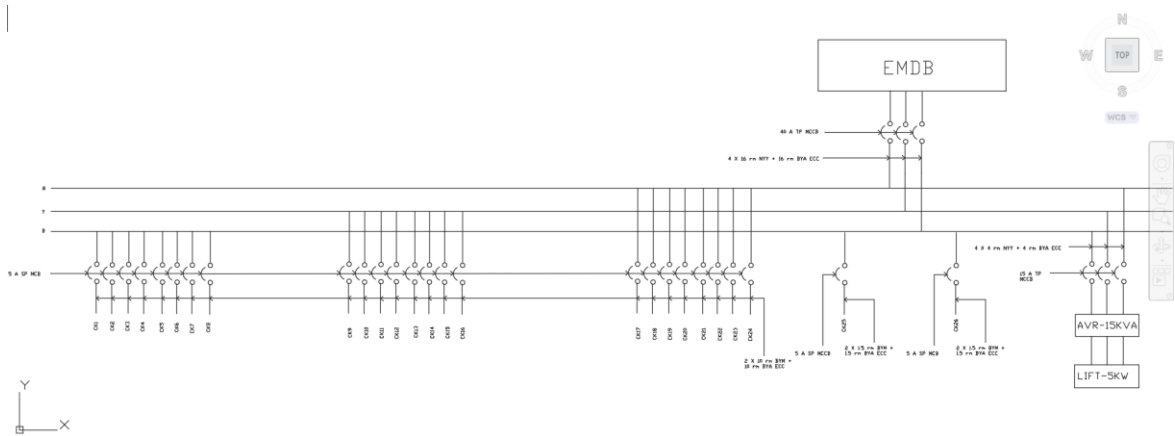
EBDB Diagram for Ground:



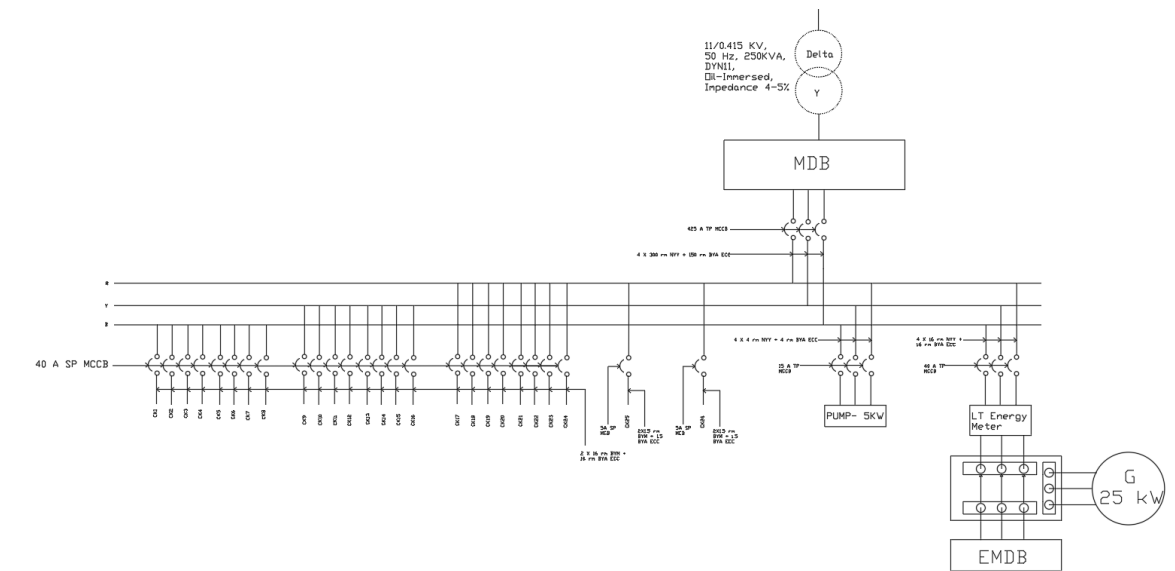
EBDB Diagram for Rooftop:



EMDB Diagram (Entirety):



MDB Diagram (Entirety)



Load Calculations:

The diagrams are shown below, and the necessary load calculations used are shown in the next section

BDB Floor Switchboard Conduits:

Circuit 1:

$$I = \frac{\text{Total Power}}{\text{Voltage} \times \text{Power factor}}$$
$$= \frac{5 \times 20 + 3 \times 100}{220 \times 0.7}$$

$$= 2.6 \text{ A}$$

2 x 1.5 cm BYB + 1.5 cm ^{ECC} BYA used

Circuit 2:

$$I = \frac{5 \times 20 + 4 \times 100}{220 \times 0.7} = 3.25 \text{ A}$$

2 x 1.5 cm BYB and 1.5 cm ECC BYA used

Circuit 3:

$$I = \frac{20 + 100}{220 \times 0.7} = 0.779 \text{ A}$$

2 x 1.5 cm BYB + 1.5 cm ECC BYA used

Ground BDB Switchboard Conduits:

Circuit 1:

$$I = \frac{12 \times 20 + 100}{220 \times 0.7} = 2.21 \text{ A}$$

2 x 1.5 cm BYB + 1.5 cm BYA ECC used

Circuit 2:

$$I = \frac{5 \times 20 + 3 \times 100}{220 \times 0.7} = 2.60 \text{ A}$$

2 x 1.5 cm BYB + 1.5 cm BYA ECC used

Roof top BDB Switchboard Conduits:

Circuit 1:

$$I = \frac{4 \times 20}{220 \times 0.7} = 0.519 \text{ A}$$

2 x 1.5 cm BYB + 1.5 cm BYA ECC used

Circuit 2:

$$I = \frac{6 \times 20}{220 \times 0.7} = 0.779 \text{ A}$$

2 x 1.5 cm BYB + 1.5 cm BYA ECC used

Circuit 3:

$$I = \frac{12 \times 20}{220 \times 0.7} = 1.56 \text{ A}$$

2 x 1.5 cm BYB + 1.5 cm BYA ECC used

Floor EBDB Switchboard Conduits:

Circuit 1:

$$I = \frac{100 + 20}{220 \times 0.7} A = 0.779 A$$

2x1.5 mm BYM + 1.5 mm BYA ECC needed

Circuit 2:

$$I = \frac{20}{220 \times 0.7} A = 0.13 A$$

2x1.5 mm BYM + 1.5 mm BYA ECC used

Circuit 3:

$$I = \frac{120}{220 \times 0.7} A = 0.779 A$$

2x1.5 mm BYM + 1.5 mm BYA ECC used

Circuit 4:

$$I = \frac{120}{220 \times 0.7} A = 0.779 A$$

2x1.5 mm BYM + 1.5 mm BYA ECC used

Circuit 5:

$$I = \frac{20}{220 \times 0.7} A = 0.13 A$$

2x1.5 mm BYM + 1.5 mm BYA ECC used

Circuit 6:

$$I = \frac{2 \times 20 + 100}{220 \times 0.7} A = 0.909 A$$

2x1.5 mm BYM + 1.5 mm BYA ECC used

Circuit 7:

$$I = \frac{20}{220 \times 0.7} A = 0.13 A$$

2x1.5 mm BYM + 1.5 mm BYA ECC used

Circuit 8:

$$I = \frac{3 \times 20 + 100}{220 \times 0.7} A = 1.04 A$$

2x1.5 mm BYM + 1.5 mm BYA ECC used

Ground EBDB Switchboard:

Box ESB3 (Circuit 1):

$$I = \frac{9 \times 20}{220 \times 0.7} A = 1.17 A$$

2x1.5 mm BYM + 1.5 mm BYA ECC used

Roof-top EBDB Switchboard:

Circuit 1:

$$I = \frac{3 \times 20}{220 \times 0.7} A = 0.39 A$$

2x1.5 mm BYM + 1.5 mm BYA ECC used

BDB Diagram Calculation:

$$\text{Total BDB load} = \text{load from circuits} \times 0.75 + \text{Total socket load} \times 0.25$$

$$\text{BDB current} = \frac{\text{Total BDB Load}}{\text{voltage} \times \text{p.f.}}; \text{Socket Load} = 4000\text{W}$$

$$\text{Voltage} = 220\text{V}, \text{p.f.} = 0.7$$

For each unit:

$$\text{Circuit 1 Load} = 400\text{W}$$

$$\text{Circuit 2 Load} = 500\text{W}$$

$$\text{Circuit 3 Load} = 120\text{W}$$

$$\text{No. of } 4000\text{W (Q) sockets} = 5$$

$$\text{Total socket load} = 4000 \times 5\text{W} = 20000\text{W}$$

$$\text{BDB current} = \frac{1020 \times 0.75 + 20000 \times 0.25}{220 \times 0.7} = 37.4\text{A}$$

So, we used 40A SP MCCB from MDB to BDB
Wires is $2 \times 16\text{mm}^2 \text{BYM} + 16\text{mm}^2 \text{BYA ECC}$

Ground:

$$\text{Circuit 1 Load} = 340\text{W}; \text{Circuit 2 Load} = 400\text{W}$$

$$\text{BDB Current} = \frac{340 \times 0.75}{220 \times 0.7} \text{A} = 3.60\text{A}$$

5A SP ^{MCCB} needed from MDB to BDB

Wires is $2 \times 1.5\text{mm}^2 \text{BYM} + 1.5\text{mm}^2 \text{BYA ECC}$

Roof top:

$$\text{Circuit 1 Load} = 80\text{W}$$

$$\text{Circuit 2 Load} = 120\text{W}$$

$$\text{BDB Current} = \frac{200 \times 0.75}{220 \times 0.7} = 0.974\text{A}$$

5A SP MCCB needed from BMD to BDB

$2 \times 1.5\text{mm}^2 \text{BYM} + 1.5\text{mm}^2 \text{BYA ECC}$

EBDB Diagram Calculation:

for each unit:

Circuit 1 Load = 120W, similarly:

Circuit 2 Load = 20W, Circuit 3 Load = 120W

Circuit 4 Load = 120W, Circuit 5 Load = 20W

Circuit 6 Load = 140W, Circuit 7 Load = 20W

Circuit 8 Load = 160W

Total Load = $(20 + 120 + 120 + 20 + 140 + 20 + 160)W$

$$= 600W \quad 0.75 \times$$

$$EBDB \text{ current} = \frac{600 \times 0.75}{220 \times 0.7} = 3.90A \quad 4.68A$$

5A SP MCB needed from EMDB to EBDB

2x1.5mm BYM + 1.5mm BYA ECC used (wire)

Floor Lobby:

Each floor power = $6 \times 20W = 120W$

$$EBDB \text{ Ground current} = \frac{120 \times 6 \times 0.75}{220 \times 0.7} A$$

$$= 3.51A$$

5A SP MCB needed from ~~EMDB~~ ground to EBDB

2x1.5mm BYM + 1.5mm BYA ECC used

Ground:

Circuit 1 Load = 180W

$$EBDB \text{ current} = \frac{180 \times 0.75}{220 \times 0.7} = 1.17A \quad 0.75 \times$$

5A SP MCB needed $= 0.878A$

wire is 2x1.5mm BYM + 1.5mm BYA ECC

from EMDB to EBDB.

Roof top:

Circuit 1 Load = 60W

$$EBDB \text{ current} = \frac{60 \times 0.75}{220 \times 0.7} = 0.39A \quad 0.75 \times$$

5A SP MCB needed from ~~EMDB~~

to EMDB to EBDB

wire is 2x1.5mm BYM + 1.5mm BYA ECC

Electric Doorbells use negligible electricity (U.F. too low), so we have neglected them in calculation.

Emergency Main Distribution Board (EMDB) Calculations:

There are 6 floors ~~for~~ with 4 residential units in each floor. Total of $6 \times 4 = 24$ units

$$\begin{aligned} \text{Total power} &= \text{Power of all floors} + \text{EBDB ground total power} + \text{EBDB total power in ground} \\ &\quad \text{(all units)} \quad \quad \quad \text{(to floor lobby)} \\ &\quad + \text{Rooftop power} + \text{Lift power (Lifts have u.f. around 0.3)} \\ &= (720 \times 0.75 \times 24 + 720 \times 0.75 + 180 \times 0.75 + 60 \times 0.75 + 5000 \times 0.3) \text{ W} \\ &= 15180 \text{ W} \end{aligned}$$

We know, for a three-phase system:

$$P = 3VI \times \text{p.f.} \Rightarrow I = \frac{15180}{3 \times 220 \times 0.7} = 32.9 \text{ A}$$

40A TP MCCB needed

40x16 mm NYY + 16 mm DIA ECC used

We used a 25kW Generator for emergency, and it is connected to the EMDB via an automatic transfer switch (ATS). Under normal operation, the MDB provides the load, whereas during emergency, the 25 kW Generator does.

Main Distribution Board (MDB) Calculations:

Total Power = Power in all units + power in ground floor + power on rooftop
+ Pump power + EMBB total power

$$= \{(1020 \times 0.75 + 20000 \times 0.25) \times 24 + 740 \times 0.75 + 200 \times 0.75 + 5000 \times 1.0 + 15180\} \text{ W}$$

$$= 159245 \text{ W}$$

$$P = 3VI \times \text{p.f.} \Rightarrow I = 344.7 \text{ A}$$

425A TP MCCB is needed from MDB to BDB

4x 300mm NYY + 150 mm BYA ECC wire needed

Calculations for Three-Pin Sockets in BDB (one unit)

For each socket, power rating = 4000W

$$I = \frac{4000}{220 \times 0.7} = 26.0 \text{ A}$$

$\therefore$ 30A SP MCB used

Wire is 2x 10mm BYM + 10mm BYA ECC

Calculation for Pump and Lift (5kW)

$$I = \frac{5000}{3 \times 220 \times 0.7} = 10.82 \text{ A}$$

$\therefore$ for both Pump and Lift, we need 15A TP MCCB
4x 4mm NYY + 4mm BYA ECC

Summary Table:

Below is the summary table, showing the number of appliances:

Name of the Appliance	Number of Appliance Used in the Entire Building	Cable and Conduits Used
Ceiling Fan	$8 \times 24 + 3 = 195$	2X1.5 rm BYM for one appliance (C1)
Bracket-Mounted Light Bulb	$15 \times 24 + 7 + 22 = 389$	2X1.5 rm BYM for one appliance (C1)
Ceiling-Mounted Light Bulb	$7 \times 24 + 5 + 1 + 20 + 1 = 195$	2X1.5 rm BYM for one appliance (C1)
2 Pin 100W Socket at SB level	$6 \times 24 + 10 + 7 = 161$	2X1.5 rm BYM for one appliance (C1)
3 Pin 20A Socket	$5 \times 24 = 120$	2X10 rm BYM + 10 rm BYA ECC for one socket (C9 in diagram since it is the highest)
Number of Doorbells	24	2X1.5 rm BYM for one appliance (C1)
Total Connected Load (EMDB)	15.18 kW	4 X 16 rm NYY + 16 rm BYA ECC
Total Connected Load (MDB)	159.245 kW	4 X 300 rm NYY + 150 rm BYA ECC

Project Summary:

In this project, we designed a seven-story residential building using AutoCAD 2027, with 6 residential floors, containing 4 units each floor. For the floor plan, we decided to have 4 living rooms and 3 toilets, along with one kitchen and one living and dining space. The living and dining space turned out to be larger than expected, requiring 3 fans. It was a bit difficult to fit them in a room, since they had chances of overlapping. Therefore, the fans were placed in a triangular formation. Even though our calculations gave us values of 1.2 fans per room sometimes, we could not place 2 fans. This is because, our width and the length of our rooms are very close, so the room is almost square-shaped. So, we had to use one fan to prevent overlap. We also connected at least one fan and light in one room to the Emergency Branch Distribution Board (EBDB). Another problem we had faced was that, we have used 4000W three-pin socket, which is seldom if ever used in residential buildings. And we also used 5 of them per unit, which caused the power demand to skyrocket. Even using a low utilization factor (since there are so many of them) did not help reduce the excessively high power demand. That's why we used a 250 kVA transformer to feed the power demand. Some compromises also had to be made in the MDB design. For example, our current was 344.7A, and we don't have wires of any such ratings, the closest we could find was 425A. This is a bit expensive due to the thickness, but it also leaves room for future expansion of the electrical system if necessary. The ground floor and the rooftop are also lit abundantly. Another problem faced was that, the table given to us has conduit ratings only up to C9. Sometimes, we connected 11, 12 appliances parallelly, in which case one conduit had to carry 24 wires. Since these ratings weren't available in the sheet provided, we named the conduits C11, C12 in those cases. Also, for the 4000W socket, we used C9, since we weren't sure of the rating and no such information was provided to us in the table.

Overall, this is a large building with a high power demand. We know that if our power demand exceeds 50 kW, we would need a separate substation. That's why, we used a dedicated 250 kVA transformer, which would be more than enough to

handle our buildings power demand of 159.245 kW. For emergency purposes, we also added a 25 kW standby generator, and we can switch between MDB and the emergency generator via an automatic transfer switch (ATS).

In conclusion, this project taught us many things about how electrical circuits are designed in a building and how the calculations are made, how the appropriate wires and conduits are chosen according to our system. This knowledge would be crucial to us in our future as electrical engineers.